

4

THIN LENSES

Diagrams of several standard forms of thin lenses were shown in Fig. 3A as illustrations of the fact that most lenses have surfaces that are spherical in form. Some surfaces are convex, others are concave, and still others are plane. When light passes through any lens, refraction at each of its surfaces contributes to its image-forming

A thin lens may be defined as one whose thickness is considered small in comparison with the distances generally associated with its optical properties. Such distances are, for example, radii of curvature of the two spherical surfaces, primary and secondary focal lengths, and object and image distances.

4.1 FOCAL POINTS AND FOCAL LENGTHS

Diagrams showing the refraction of light by an equiconvex lens and by an equiconcave lens are given in Fig. 4A. The axis in each case is a straight line through the geometrical center of the lens and perpendicular to the two faces at the points of intersection. For spherical lenses this line joins the centers of curvature of the two surfaces. The primary focal point F is an axial point having the property that any ray coming from it or proceeding toward it travels parallel to the axis after refraction.

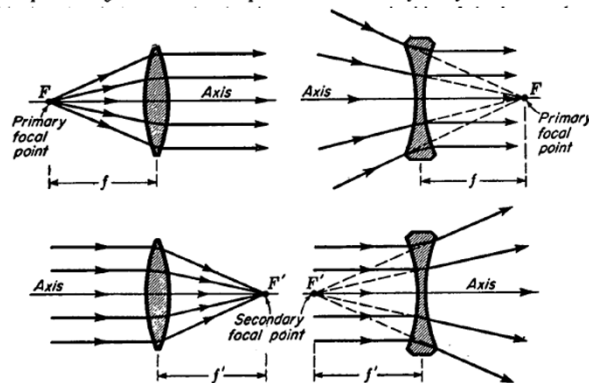


FIGURE 4A
Ray diagrams illustrating the primary and secondary focal points F and F' and the corresponding focal lengths f and f' of thin lenses.

Every thin lens in air has two focal points, one on each side of the lens and equidistant from the center. This can be seen by symmetry in the cases of equiconvex and equiconcave lenses, but it is also true for other forms provided the lenses may be regarded as thin. *The secondary focal point F' is an axial point having the property that any incident ray traveling parallel to the axis will, after refraction, proceed toward, or appear to come from, F' .* The two lower diagrams in Fig. 4A are given for the purpose of illustrating this definition. In analogy to the case of a single spherical surface (see Chap. 3), a plane perpendicular to the axis and passing through a focal point is called a *focal plane*. The significance of the focal plane is illustrated for a converging lens in Fig. 4B. Parallel incident rays making an angle θ with the axis are brought to a focus at a point Q' in line with the chief ray. The chief ray in this case is defined as the ray which passes through the center of the lens.

$$f = f'$$

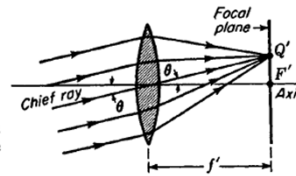


FIGURE 4B
How parallel incident rays are brought to a focus at the secondary focal plane of a thin lens.

4.2 IMAGE FORMATION

When an object is placed on one side or the other of a converging lens and beyond the focal plane, an image is formed on the opposite side (see Fig. 4C). If the object is moved closer to the primary focal plane, the image will be formed farther away from the secondary focal plane and will be larger, i.e., magnified. If the object is moved farther away from F , the image will be formed closer to F' and will be smaller.

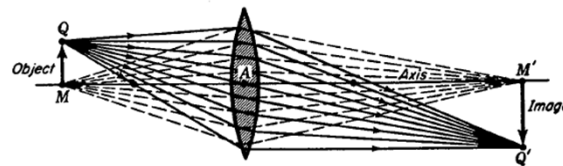


FIGURE 4C
Image formation by an ideal thin lens. All rays from an object point Q which pass through the lens are refracted to pass through the image point Q' .

4.3 CONJUGATE POINTS AND PLANES

If the principle of the reversibility of light rays is applied to Fig. 4C, we observe that $Q'M'$ becomes the object and QM becomes its image. The object and image are therefore *conjugate*, just as they are for a single spherical surface (see Sec. 3.4). Any pair of object and image points such as M and M' in Fig. 4C are called *conjugate points*, and planes through these points perpendicular to the axis are called *conjugate planes*.

If we know the focal length of a thin lens and the position of an object, there are three methods of determining the position of the image: (1) graphical construction, (2) experiment, and (3) use of the lens formula

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad (4a)$$

Here s is the object distance, s' is the image distance, and f is the focal length, all measured to or from the center of the lens.

4.4 THE PARALLEL-RAY METHOD

The parallel-ray method is illustrated in Fig. 4D. Consider the light emitted from the extreme point Q on the object. Of the rays emanating from this point in different directions the one (QT) traveling parallel to the axis will by definition of the focal point

be refracted to pass through F' . The ray QA , which goes through the lens center where the faces are parallel, is undeviated and meets the other ray at some point Q' . These two rays are sufficient to locate the tip of the image at Q' , and the rest of the image lies in the conjugate plane through this point. All other rays from Q will also be brought to a focus at Q' . As a check, we note that the ray QF which passes through the primary focal point will by definition of F be refracted parallel to the axis and will cross the others at Q' as shown in the figure. The numbers 1, 2, 3, etc., in Fig. 4D indicate the order in which the lines are customarily drawn.

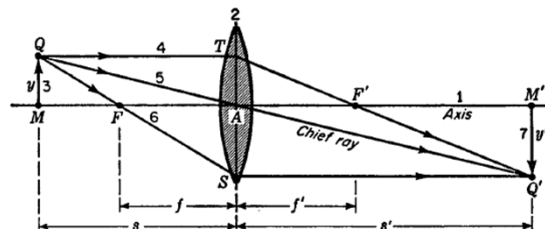


FIGURE 4D
The parallel-ray method for graphically locating the image formed by a thin lens.

4.5 THE OBLIQUE-RAY METHOD

Let MT in Fig. 4E represent any ray incident on the lens from the left. It is refracted in the direction TX and crosses the axis at M' . The point X is located at the intersection between the secondary focal plane $F'W$ and the dashed line RR' drawn through the center of the lens parallel to MT .

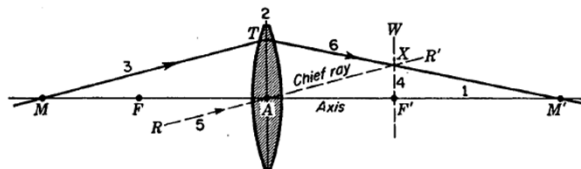


FIGURE 4E
The oblique-ray method for graphically locating the image formed by a thin lens.

4.6 USE OF THE LENS FORMULA

$$s' = \frac{s \times f}{s - f} \quad (4b)$$

The sign conventions to be used for the thin-lens formulas are identical to those for a single spherical surface given in Sec. 3.5.

4.7 LATERAL MAGNIFICATION

A simple formula for the image magnification produced by a single lens can be derived from the geometry of Fig. 4D. By construction it is seen that the right triangles QMA and $Q'M'A$ are similar. Corresponding sides are therefore proportional to each other, so that

$$\frac{M'Q'}{MQ} = \frac{AM'}{AM}$$

where AM' is the image distance s' and AM is the object distance s . Taking upward directions as positive, $y = MQ$, and $y' = -M'Q'$; so we have by direct substitution $y'/y = -s'/s$. The lateral magnification is therefore

$$m = \frac{y'}{y} = -\frac{s'}{s} \quad (4c)$$

When s and s' are both positive, as in Fig. 4D, the negative sign of the magnification signifies an inverted image.

4.8 VIRTUAL IMAGES

The images formed by the converging lenses in Figs. 4C and 4D are real in that they can be made visible on a screen. They are characterized by the fact that rays of light are actually brought to a focus in the plane of the image. A virtual image cannot be formed on a screen (see Sec. 3.3). The rays from a given point on the object do not actually come together at the corresponding point in the image; instead they must be projected backward to find this point. Virtual images are produced with converging lenses when the object is placed between the focal point and the lens and with diverging lenses when the object is in any position. Examples are shown in Figs. 4F and 4G.

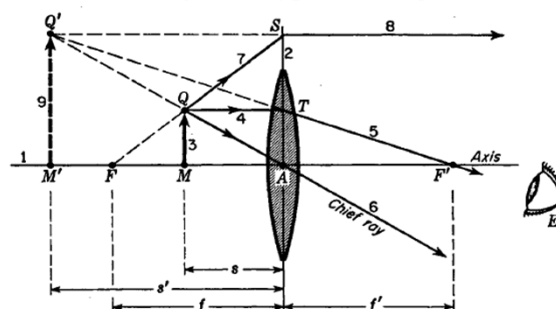


FIGURE 4F

The parallel-ray method for graphically locating the virtual image formed by a positive lens when the object is between the primary focal point and the lens.

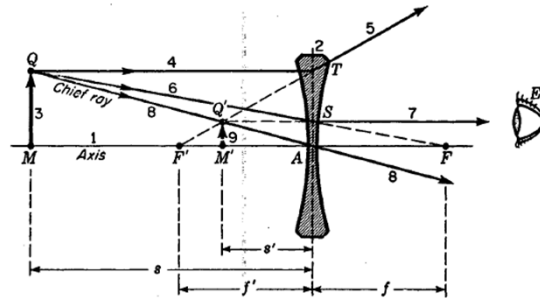


FIGURE 4G
The parallel-ray method for graphically locating the virtual image formed by a negative lens.

4.9 LENS MAKERS' FORMULA

If a lens is to be ground to some specified focal length, the refractive index of the glass must be known. It is customary for manufacturers of optical glass to specify the refractive index for yellow sodium light, the D line. Supposing the index to be known, the radii of curvature must be so chosen as to satisfy the equation

$$\bullet \quad \frac{1}{f} = (n - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (4d)$$

As the rays travel from left to right through a lens, *all convex surfaces are taken as having a positive radius and all concave surfaces a negative radius*. For an equiconvex lens like the one in Fig. 3A(a), r_1 for the first surface is positive and r_2 for the second surface negative. Substituting the value of $1/f$ from Eq. (4a), we write

$$\frac{1}{s} + \frac{1}{s'} = (n - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (4e)$$

4.10 THIN-LENS COMBINATIONS

The principles of image formation presented in the preceding sections of this chapter are readily extended to optical systems involving two or more thin lenses. Consider, for example, two converging lenses spaced some distance apart as shown in Fig. 4H. Here an object Q_1M_1 is located at a given distance s_1 in front of the first lens, and an image $Q_2'M_2$ is formed some unknown distance s_2' from the second lens. We first apply the graphical methods to find this image distance and then show how to calculate it by the use of the thin-lens formula.

The first step in applying the graphical method is to disregard the presence of the second lens and find the image produced by the first lens alone. In the diagram the parallel-ray method, as applied to the object point Q_1 , locates a real and inverted image at Q_1' . Any two of the three incident rays 3, 5, and 6 are sufficient for this purpose. Once Q_1' is located, we know that all the rays leaving Q_1 will, upon refraction by the first lens, be directed toward Q_1' . Making use of this fact, we construct a fourth ray by drawing line 9 back from Q_1' through A_2 to W . Line 10 is then drawn in connecting W and Q_1 .

The second step is to imagine the second lens in place and to make the following changes. Since ray 9 is seen to pass through the center of lens 2, it will emerge without deviation from its previous direction. Since ray 7 between the lenses is parallel to the axis, it will upon refraction by the second lens pass through its secondary focal point F_2' . The intersection of rays 9 and 11 locates the final image point Q_2' . Q_1 and Q_1' are conjugate points for the first lens, Q_2 and Q_2' are conjugate points for the second lens, and Q_1 and Q_2' are conjugate for the combination of lenses. When the image $Q_2'M_2$ is drawn in, corresponding pairs of conjugate points on the axis are M_1 and M_1' , M_2 and M_2' , and M_1 and M_2' .

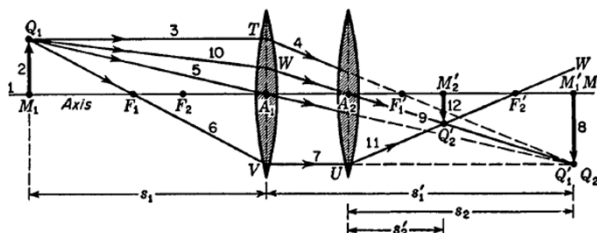


FIGURE 4H
The parallel-ray method for graphically locating the final image formed by two thin lenses.

The oblique-ray method given in Fig. 4E is applied to the same two lenses in Fig. 4I. A single ray is traced from the object point M to the final image point M'_2 . The lines are drawn in the order indicated. The dotted line 6 is drawn through A_1 parallel to ray 4 to locate the point R'_1 . The dotted line 9 is drawn through A_2 parallel to ray 7 to locate R'_2 . This construction gives the same conjugate points along the axis. Note that the axis itself is considered as the second light ray in locating the image point M'_2 .

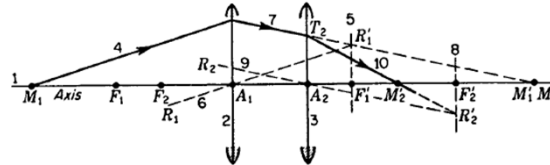


FIGURE 4I
The oblique-ray method for graphically locating the final image formed by two thin lenses.

4.11 OBJECT SPACE AND IMAGE SPACE

For every position of the object there is a corresponding position for the image. Since the image may be either real or virtual and may lie on either side of the lens, the *image space* extends from infinity in one direction to infinity in the other. But object and image points are conjugate; so the same argument holds for the *object space*. In view of their complete overlapping, one might wonder how it is that the distinction between object and image space is made. This is done by defining everything that pertains to the rays before they have passed through the refracting system as belonging to the object space and everything that pertains to them afterward as belonging to the image space. Referring to Fig. 4H, the object Q_1 and the rays Q_1T , Q_1A_1 , and Q_1V are all in the object space for the first lens. Once these rays leave that lens, they are in the image space of the first lens, as is also the image Q'_1 . This space is also the object space for the second lens. Once the rays leave the second lens, they and the image point Q'_2 are in the image space of the second lens.

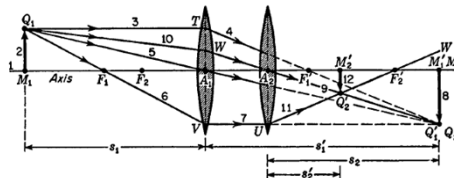


FIGURE 4H
The parallel-ray method for graphically locating the final image formed by two thin lenses.

4.12 THE POWER OF A THIN LENS

The concept and measurement of lens power correspond to those used in the treatment of reduced vergence and the power of a single surface as given in Sec. 3.9. The power of a thin lens in diopters is given by the reciprocal of the focal length in meters:

$$\bullet \quad P = \frac{1}{f} \quad \text{diopters} = \frac{1}{\text{focal length, m}} \quad (4f)$$

By making use of the lens makers' formula [Eq. (4d)] we can write

$$P = (n - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (4g)$$

where r_1 and r_2 are the two radii, measured in meters, and n is the refractive index of the glass.

4.13 THIN LENSES IN CONTACT

When two thin lenses are placed in contact, as shown in Fig. 4J, the combination will act as a single lens with two focal points symmetrically located at F and F' on opposite sides. Parallel incoming rays are shown refracted by the first lens toward its secondary focal point F'_1 . Further refraction by the second lens brings the rays together at F' . This latter is defined as the secondary focal point of the combination, and its distance from the center is defined as the combination's secondary focal length f' .

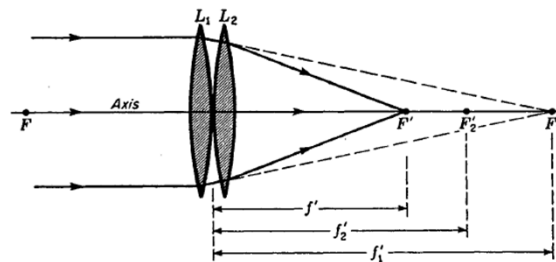


FIGURE 4J

The power of a combination of thin lenses in contact is equal to the sum of the powers of the individual lenses.

If we now apply the simple lens formula (4a) to the rays as they enter and leave the second lens L_2 , we note that for the second lens alone f'_1 is the object distance (taken with a negative sign), f' is the image distance, and f'_2 is the focal length. When Eq. (4a) is applied, these substitutions for s , s' , and f , respectively, give

$$\frac{1}{-f'_1} + \frac{1}{f'} = \frac{1}{f'_2} \quad \text{or} \quad \frac{1}{f'} = \frac{1}{f'_1} + \frac{1}{f'_2}$$

Since we have assumed that the lenses are in air, the primary focal lengths are all equal to their respective secondary focal lengths and we can drop all primes and write

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \quad (4h)$$

we can write $P_1 = 1/f_1$, $P_2 = 1/f_2$, and $P = 1/f$, we obtain for the power of the combination

$$P = P_1 + P_2 \quad (4i)$$

4.14 DERIVATION OF THE LENS FORMULA

Homework:

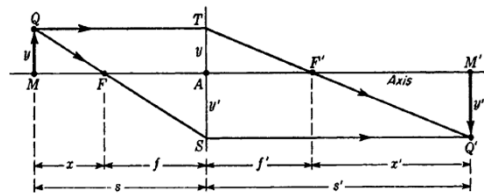


FIGURE 4K
The geometry used for the derivation of thin-lens formulas.

4.15 DERIVATION OF THE LENS MAKERS' FORMULA

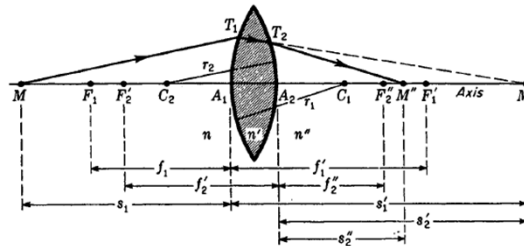


FIGURE 4L
Each surface of a thin lens has its own focal points and focal lengths, as well as separate object and image distances.